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Title : Diagram Explanation

Full : Diagram : Explanation :

- Either we use a laptop or mobile phone for a client application where users access it
- Client is a service or a program which meets the server data and browser is a another single client
- A client can be a browser , curl , postman or Another Web Application
- A curl is basically command line utility such as curl https://google.com and it is application running in my machine and its an another web application where it understands web protocol all we need to know its knows http language and speak http language
- There is a different OS and different written SDK and what we do is to start with a programming language as Java ,Python , JavaScript to create a web application
- To create web application it should have libraries , frameworks or SDKs
- SDK known as Software Development Kit
- This talks to OS and Network and basically when writing a web application it starts with choosing programming language which is the first layer
- Those programming languages have multiple libraries and frameworks and Java as known as spring boot which is framework and in Python have Fast API
- Each language has a framework and even building or tools even a command line utility and each should have libraries , framework , SDKs
- These all will be available in open source
- Open source known as free sources available to you which you can view or download by any users
- When running we do compile , interpret or package it and Java Complies and Python does not compile and Python interpret
- Compile means you convert a programming language you understand like human readable format into machine readable format
- Transferring machine readable format is known as compile
- Java is complier and Python is interpreter

- We use libraries for complex programs and not for simple code : Hello World !
- Source file are just text file not executable and does not run and its just a input . To run we either compile or interpret
- Our system like laptop or mobile should have a runtime environment for either Java or Python to interpret or compile
- There two components while installing such as developers or runtime and runtime environments such as JRE for Java
- Runtime converts to machine language which is used for
- Runtime will interact with OS with drivers and while running code it should be in RAM and not run in disks such hard disk for source code
- We have severs as single server or multiple server and all servers will run the same code and there will be some entity as load balancer
- Load balancer will have multiple instances and load means any traffic comes to the server is called load
- Load balancer will spilt traffic into multiple servers
- Three servers are planned and all three have spiked traffic and they use scaling method with manage traffic
- Scaling means to expand if more traffic and reduce if less traffic
- HTTPS and HTTP are used for Web Traffic
- Same traffic as HTTPS and HTTP are split into different servers by a common load balancer and not changing the protocol
- We deploy the same with the servers such as Programming Language , Libraries , Framework and SDK followed by Compile , Interpret , and packing and giving runtime through OS such as drivers using network and RAM
- Client software will be available for user end such as developers and having an application where anyone can use it
- Client actually understands HTTP Protocols and want something data from the server and client application is actually being served by the server
- These traffic are sent to the data transfer such as HTML , CSS , JavaScript and given through Databases such as SQL and etc to store data in rows and columns for complex data
- Images and videos or backups are the files used to store unstructured data using Object Storage as AWS S3 and Blob Storage

Therefore How the whole requests works :

- Request starts at the client and Load Balancer decides which server is least busy and server processes the logic using RAM
- Server asks to find an database available for storing data and database send the data backup to the user

Some Basic Concepts based on the diagram :

The primary function of the Load Balancer is :

- To split incoming traffic to various server based on specific rules
- Load Balancer states that the Application Load Balancer splits traffic based on rules to manage the load across multiple servers

In server what does scaling horizontal used as :

- It increases the number of servers to handle more requests

The protocol used for communication between the laptop or mobile client and the load balancer :

- Using HTTPS / HTTP
- Connecting the clients to the Application load balancer with HTTP and HTTPS with standard web protocols

The unstructured data like images or videos are stored in a architecture as :

- It is stored in AWS S3 or Blob storage
- These are under data storage which are designed for binary files like images and videos

Java , Python , JavaScript as the server plays which role in the architecture :

- These are languages used to write the backend application logic
- Server layer uses these languages and runtimes to process data and handle requests

In server OS/Drivers are based in the layer :

- A low level instruction set and operating systems that manage hardware
- This layer represents connection between application code and physical server hardware